

My work focuses on problems in machine learning theory and graph theory. The main theme of my research is to **understand the dynamics of information flow in learning and networked systems**. I have a broad interest in ML and networks, and in particular I have spent the last two years of my postdoc appointment studying the foundations of LLMs [16] and how communication network structure impacts data integrity [20, 18, 22, 21, 26, 25, 27]. I have worked on models for DNA data storage [19], graph-based private data retrieval [23], and interactive graph learning [11]. I develop and utilize tools from combinatorics, information theory, and dynamics to tackle interesting problems at the intersection of computer science, mathematics, and engineering. The next three sections outline the problems and techniques I have developed over my career, with particular focus on the following questions,

1. What are the foundations of self-attention in large language models? I view self-attention as an interacting particle system [16], where information flow is captured by token dynamics. I am interested in dimensionality reduction techniques and conservation laws that make these systems mathematically interpretable. This is the primary focus for my current and future work.
2. How does network topology affect data quality? I have answered this question for the *gossip* communication model [18, 20, 26, 27] by leveraging tools from graph theory and statistical physics. Here packets traversing a communication network form the information flow.
3. How many imperfect observations are needed to recover a combinatorial object? I am interested in algorithms and lower bounds for two concrete models of this question: reconstructing DNA from subsequences [19] and reconstructing automata from random walks [17]. In both models, information flows to the learner via noisy observations.

1 Large Language Models as Interacting Particle Systems

During my PhD I focused on classical notions of learning (PAC learning, query learning, online learning, etc.). Modern deep-learning algorithms, including LLMs, are very difficult to study under these frameworks since gradient descent on the loss landscape is a black box. To address this limitation, a recent line of work has formalized the inference-time behavior of LLMs from an interacting particle perspective [12]. Given the recent surge of LLM use in STEM research, this direction is very timely and provides a mathematical framework to explain these systems. The key feature that separates LLMs from other neural networks is the *self-attention layer*, which allows the LLM to extract long-distance correlations in natural language. It was recently shown that application of repeated self-attention layers is modeled, in an appropriate sense, by a nonlinear ordinary differential equation (ODE) which is driven by the high-dimensional learned parameter matrices [12, 3]. The equation has the form

$$\dot{x}_k(t) = P_{x_k} \left(f_{A,V}(x_k(t), \{x_i(t)\}_{i=1}^n) \right), \quad (1)$$

where $x_k \in \mathbb{S}^{d-1}$ represents a token on the unit sphere, P_{x_k} projects onto the tangent space of x_k , and f is a function that depends on the location of token x_k , the collective locations of all particles $\{x_i\}_{i=1}^n$, and two $d \times d$ parameter matrices A and V . Along with a number of collaborators, I have shown that linear self-attention¹ with $f = \frac{1}{n} \sum_{i=1}^n \langle x_i, Ax_k \rangle V x_k$ in $d = 2$ is in fact a Kuramoto-Daido model [16]. The Kuramoto model is a classical system used to explore phase transitions, synchronization, and chaos in populations of coupled agents [31] (you may have seen demonstrations of pendulums synchronizing on a wobbly table; this is the physical motivation for Kuramoto). The connection between self-attention and the Kuramoto model allows us to leverage powerful dimensionality reduction techniques, **explaining the long-time system behavior for many classes of parameter matrices**. Previous work could only describe the long-time behavior for a very restricted class of parameters by applying gradient flow techniques [12, 6, 3]. LLMs work on a sequence of N tokens embedded in $\mathbb{R}^d$, leading to a coupled ODE in Nd variables. For $d = 2$ our study shows linear self-attention is essentially governed by a single complex-valued ODE — such an immense reduction in variables allows us to rigorously describe the long-time behavior of the system for a wide range of parameters, even as $N \rightarrow \infty$. Our reduction generalizes to arbitrary dimensions, which is the topic of a

¹Here ‘linear’ refers to the self-attention kernel used inside f , the underlying ODE/PDE is still nonlinear.

follow-up study by one of our collaborators where I am conducting analysis on a broad range of parameter matrices. The idea for the reductions is that linear attention evolves tokens via a *pure second-harmonic coupling*, which preserves certain group structures forcing low-dimensional dynamics.

Our paper also shows linear self-attention in $d = 2$ can display a wide range of dynamical behavior in the long-time limit. We have identified classes of A and V parameters which force tokens to converge to a pair of antipodal points $\pm\theta_\infty \in \mathbb{S}^1$, cluster to a constant velocity antipodal pair, partially cluster, and neither cluster nor converge. Long-time behavior is related to the expressability of LLMs, so showing such rich behavior already in $d = 2$ demonstrates the power of LLMs and gives evidence why analysis in $d > 2$ for arbitrary A and V has proven so difficult. The analysis of the cases above is aided by numerical software that I wrote for identifying the dynamics based on phase portraits of the reduced dynamics. Additionally, I have identified that there are parameter regimes where the system evolves as a **Hamiltonian flow**, which is surprising given that previous work has focused heavily on gradient flow methodologies. This opens a new research direction: identify parameter regimes of self-attention which admit energy conservation instead of energy minimization. In addition, I am studying the interplay of A and V in large embedding dimension, as well as analyzing the role of the feed-forward layer using control-theoretic tools.

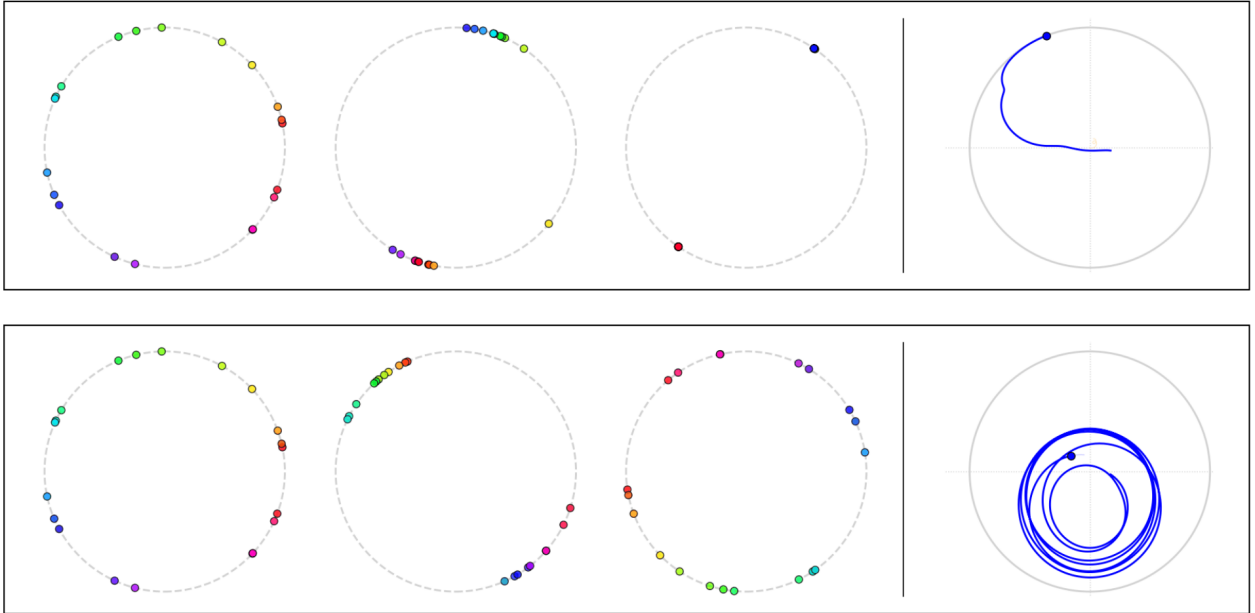


Figure 1: Snapshots of two systems with 20 particles evolving under linear attention with different choices of the matrices A and V . The left three columns show the particle positions at three time points, and the right-most column shows the corresponding time evolution of the so-called *order parameter* which measures token synchrony. In the system at the top, the dynamics form two antipodal clusters, and the associated order parameter converges. In the system at the bottom, the particles and order parameter do not converge. Our work described above [16] rigorously analyzes the long-time behavior of such systems.

2 Gossip Networks

One of the simplest communication protocols used in real systems is known as *gossip*, in which each user randomly sends and receives updates from nearby users. This protocol uses very limited computational resources at each user, since it only requires local information of the underlying graph structure. A fundamental question for this local communication model is: how new is the information held by a given user at a particular moment? There is necessarily a tradeoff between communication overhead and information quality in networked systems; centralized protocols have high communication overhead but good information quality, while local protocols have the opposite tradeoff. The *age of information* (AoI) [15] is one measure

of information quality that has seen a surge of interest in the information theory community. Informally the AoI of a packet of data measures how out of date the packet is by measuring the elapsed time since the packet’s creation.

The model I consider represents the network as a graph where vertices exchange packets continuously in time according to independent Poisson processes [32, 33]. As vertices share data, they only store the most up-to-date data they have received so far. This ensures that the whole network is essentially a sensor for the ‘state of the world’, where separate nodes have imperfect but correlated tracking of the world. My goal is to understand how the overall network structure impacts the quality of sensing as measured by the AoI metric. This complements other work, which optimizes the communication model while keeping the network structure fixed [24].

Combinatorial Approaches

The first works in the area considered the long-time average age in the complete graph and cycle graph [32, 33]. The proof technique utilized tools from control theory to express the AoI as a certain recursion on subsets of finite graphs. **I strengthened this foundational result** [22] by giving a simpler proof with no reference to control, allowing these recursions to hold on locally finite infinite graphs. The recursive formula is the starting point for many works in age of gossip [29, 5]. The main technical challenge for finding the AoI is that the recursion tree is exponential in size of the graph. I became interested in AoI when I attended a talk by Srivastava and Ulukus [28] where they bounded the recursion using properties of the 2-D square grid. Their proof contained hints of a connection between AoI and the connectivity of the graph. As a PhD student **I identified the Cheeger constant as a combinatorial measure** for AoI in many classes of graphs [18], including random regular graphs. This led me to obtain tight bounds on the AoI in new graph families. In the same paper, I proved AoI is a monotone graph property (solving an open problem and justifying earlier results [30, 14]) and leveraged this fact to **prove a thresholding phenomena in the Erdős–Rényi random graph**. I have also identified graphs where small changes in the gossip protocol lead to large changes in the AoI [25].

Probabilistic Approaches and Dynamic Networks

Toward the end of my PhD I became interested in universal bounds for AoI — at that point no such result existed. My work [20] obtains such a bound, which relies on the *local structure* of the graph, as opposed to the global structure the Cheeger constant captures. We proved a connection between AoI and the statistical physics model ‘first-passage percolation’ (FPP), which allowed us to **develop a novel combinatorial parameter tightly capturing the AoI in arbitrary graphs**. We leveraged this new connection to **solve numerous open problems** in the area. Notably the survey [13] gave the following list of graphs with unknown AoI scaling that we were able to prove tight bounds for: d -dimensional square lattices, random geometric graphs, random regular graphs, and a cycle + random matching. Additionally we found a family of graphs with arbitrary polynomial AoI scaling. In a followup work [22] I used the connection to FPP to recover the recursive identity in [32] with a very short proof and no control theoretic machinery. In particular **my derivation holds in the more general setting** of infinite, locally finite graphs and the time limit can be replaced with a suitably large stochastic time. This new probabilistic framework allowed me to study AoI in time-varying networks [27, 26, 21].

3 Reconstruction Problems

I have a general interest in problems of the form ‘how many noisy observations of a combinatorial object are necessary or sufficient to recover the original object?’, which are discussed below.

String and Tree Reconstruction

The string reconstruction problem was introduced over 20 years ago as a simplified model for DNA data storage [2]. An arbitrary binary string S is passed through a deletion channel where each bit is removed with probability p , and we call a (random) subsequence from this process a *trace* of S . The string reconstruction problem asks ‘how many traces are needed to recover an arbitrary string S with success probability $1 - \delta$?’ This is a notoriously hard problem, with the best lower bound requiring $\tilde{\Omega}\left(n^{\frac{5}{4}}\right)$ traces [7], while a recent breakthrough shows quasi-polynomially many traces suffice [4]. A similar problem was asked in the case of

trees [8], and I have proven reductions from tree reconstruction to string reconstruction. My first contribution was to show that **tree reconstruction is no easier than string reconstruction** [19]. This is somewhat surprising since trees have richer structure, we may have hoped to use their structure to obtain better sample complexity bounds. The idea of my reduction was to encode strings as a special type of tree, and show that the tree deletion channel acts essentially as the string channel on these special trees. As a converse, I showed there is a labeling of the vertices of any tree that allow it to be recovered via a coupling to the string reconstruction problem. Therefore, the problem of recovering labeled trees from traces is equivalent to the string reconstruction problem up to polynomial factors. Finally, I proved **identities for the deletion channel** itself, including a connection between bit probabilities and generating functions induced by strings.

Learning Finite Automata from Random Walks

Learning deterministic finite automata (DFA) has a rich history tracing back to the early days of learning theory as a model for autonomous robot navigation [1]. Early hardness results for passive learning were complemented by polynomial sample complexity bounds for active and query learning models. I am interested in a generalization of a learning model proposed by Freund et al. [9]. In the model I consider, an agent performs a random walk on a DFA with random transitions, and random 0-1 labels on each state. The goal of the learning procedure is to eventually make no mistakes when predicting the state labels of the DFA. In the case of an unbiased random walk, [9] proved there is an algorithm which stops making mistakes after $O(n^5)$ steps in the random walk. It turns out that their algorithm does not generalize to biased random walks. A significant portion of my PhD thesis [17] is dedicated to constructing a new algorithm which learns when the random walk is extremely biased toward taking 1-transitions. The main challenge in this setting is that very rare events contribute a significant number of errors in any algorithm. I have **constructed an algorithm that learns the core of the DFA**, making polynomial mistakes via an agnostic learning procedure. There are uncommon but not very rare events which give the learner useful information that can be leveraged to build a representation of the most visited sites in the DFA.

Future Work

My current research program is focused on the mathematics of LLMs, in particular from the interacting-particle perspective. My work provides a new framework for studying self-attention via dimensionality reduction techniques which reveal underlying symmetries in the system. I will continue studying the symmetries of self-attention along three complimentary directions. First, I will develop a general theory of symmetries and energy conservation laws in LLM self-attention dynamics. This will involve identifying invariants in the dynamics for classes of parameters A and V , for both linear self-attention and softmax attention. Second, I will characterize chaotic dynamics that arise in the self-attention model and investigate chaos in real systems. Third, I will study the dynamical role of feed-forward layers using control-theoretic methods. ***This program is inherently multi-disciplinary***; I envision students in my lab with backgrounds across computer science, mathematics, statistics, and physics.

1. Developing a theory of symmetries and conservation laws in self-attention dynamics. The first stage of this direction will identify the structural conditions that separate dissipative, conservative, and mixed dynamics. Besides our work [16], the existing literature focuses on self-attention dynamics within a gradient flow framework. This is a very restrictive assumption since trained matrices do not generally provide a gradient flow structure. To justify this research path, I have work in preparation proving Hamiltonian (i.e. energy-conserving) systems can arise in linear self-attention, which complements the existing gradient flow framework. It is currently unclear how varying the system parameters A and V break conservative or dissipative formulations, but *it is clear* that symmetry-breaking occurs. I will identify structural properties of A and V that lead to dynamics in the classes above, and I aim to find a framework that describes the long-time behavior of the dynamics for a broad class of parameters. In addition, real systems have layer-dependent A and V , so a more realistic model has time-varying parameters $A(t)$ and $V(t)$. This setting is much more complicated, as any energy landscape will itself evolve over time. This will require analysis of time-varying Riemannian metrics and analyzing flows on such manifolds.

2. Emergence of chaotic dynamics.

I have empirically observed bifurcation diagrams (Figure 2) in small test systems for both linear and

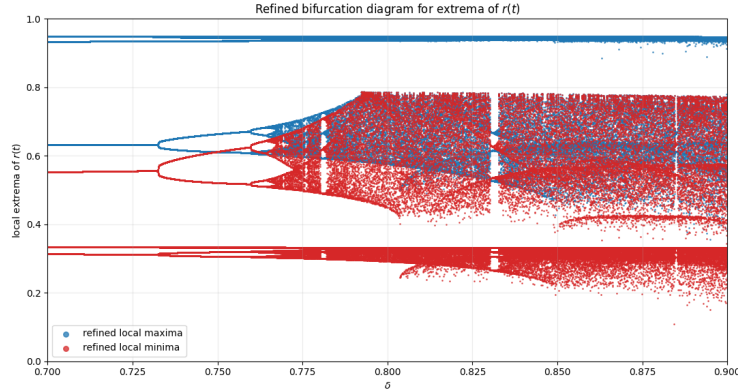


Figure 2: A bifurcation diagram illustrating the local maxima (blue) and minima (red) of the order parameter for $n = 3$ in $d = 2$ for the matrices $A = \begin{pmatrix} \delta & 0 \\ -2 & 1 \end{pmatrix}$, $V = \begin{pmatrix} \delta & 1 \\ -2 & 0 \end{pmatrix}$ in softmax attention. Note that as δ is varied between 0.7 and 0.9, a period-doubling cascade into an apparent chaotic region is observed for extrema of the order parameter. Bifurcation diagrams like this are very suggestive that the system exhibits chaos.

softmax attention dynamics, which is preliminary evidence that chaos is possible in the self-attention dynamics. The implication of this is that small changes in the prompting of an LLM can lead to very different next-token generation, which is an undesirable property from an interpretability and AI safety perspective. The goal of this direction is to identify such chaotic regimes in the idealized model, and investigate how chaos manifests in the token-level dynamics. I will characterize the invariant sets supporting the observed dynamics and determine whether they are periodic, quasiperiodic, or chaotic. In dissipative regimes this includes identifying attractors and their geometry, while in conservative regimes it includes describing invariant level sets and chaotic regions. Identifying attractors and their geometric properties for various A and V is a top priority for this direction. This compliments the study of symmetry in the dynamics as chaotic systems possess rich structure. I anticipate this starting as a numerical research direction, which is very amenable to study by first-year graduate students. Once we have compelling evidence of chaos in the self-attention dynamics, my lab will investigate if chaos appears in real LLMs, and if this chaos impacts next-token prediction.

3. Feed-forward layers as a learnable control.

As a simplifying assumption, the self-attention equation (1) does not include feed-forward layers. These layers are not only present in real LLMs, but are a key ingredient in proving that LLMs are universal approximators for maps between probability distributions [10]. Given my goal to understand the inference-time behavior of LLMs, it is important to include feed-forward layers in the model. Ultimately this leads to a system of the form:

$$\dot{x}_k(t) = P_{x_k} \left(f_{A,V} \left(x_k(t), \{x_i(t)\}_1^n \right) \right) + P_{x_k}(u_k(t)), \quad (2)$$

where the term $u_k(t)$ represents the feed-forward layer. We can view the feed-forward layer as an external forcing that acts on all particles, and only depends on their position. In particular, since feed-forward layers are themselves universal approximators we can view u_k as an external control which drives tokens in parallel at each layer with the autonomous self-attention term f . This is a useful perspective as it allows me to apply tools from continuous control theory. I have work in preparation that quantifies the tradeoff between self-attention and feed-forward layers in driving the tokens to synchronize in the long-time limit. In the same work, we have empirical evidence that real systems typically have feed-forward layers that only provide a modest external forcing. This observation leads to the question: if $\|u_k\| \ll 1$, do the dynamics (2) remain close (in an appropriate sense) to the dynamics (1), and if not are the long-time behaviors of the two systems the same? Such a stability result, along with our observation that real systems obey the hypothesis, would give further justification to study the pure self-attention dynamics. I anticipate the answer to such a question depends on the structure of A and V , which in turn depends on my findings in the first two directions above.

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